



IS5542: GenAI in Business

Abstract:

This Master's-Level course explores the transformative potential of Generative Artificial Intelligence (GenAI) in the business landscape. Students will learn core concepts, applications, and implications of GenAI across various business functions. They will gain a comprehensive understanding of how GenAI is revolutionizing industries, driving innovation, and reshaping the future of work. Through a blend of theoretical frameworks, real-world case studies, and hands-on projects, students will acquire the knowledge and skills necessary to leverage GenAI for strategic advantage and responsible implementation in their respective fields.

Learning Objectives:

Upon successful completion of this course, students will be able to:

- **Understand the fundamentals of GenAI:** Define GenAI, differentiate it from other forms of AI, and grasp its underlying technologies (e.g., large language models, deep learning).
- **Analyze GenAI applications across business functions:** Evaluate the use cases of GenAI in areas like marketing, customer service, product development, finance, and human resources.
- **Assess the impact of GenAI on business processes:** Identify opportunities for automation, efficiency gains, and innovation driven by GenAI.
- **Evaluate the ethical and societal implications of GenAI:** Discuss responsible AI development, bias mitigation, data privacy, and the potential impact on the workforce.
- **Develop strategic recommendations for GenAI adoption:** Formulate strategies for integrating GenAI into existing business models and processes.
- **Apply GenAI tools and techniques:** Gain hands-on experience with popular GenAI tools and platforms for specific business applications.

Course Content Outline (13 weeks, 3 hours per week):

Module 1: Introduction to Generative AI (2 – 3 weeks)

- Overview of GenAI key concepts and technologies
- Understanding large language models and how to evaluate LLMs' different strengths and weaknesses
- Large language models and their applications in business
- Limitations and challenges of using generative AI techniques in business contexts

Module 2: Strategic Implementation of GenAI (2 – 3 weeks)

- Identifying opportunities for GenAI adoption in organizations
- Building a GenAI strategy and roadmap for implementation
- Change management and organizational readiness for GenAI
- Evaluation metrics, Operation Plan, and Quality control

Module 3: GenAI Industry Use Cases (4 weeks)

- Marketing professionals using GenAI
- GenAI for Internal Efficiency Optimization
- GenAI in Financial Services and Insurance
- GenAI in the Retail Sector
- Guest speakers: present innovative industry practices and use cases

Module 4: Responsible AI and Future Trends (2 weeks)

- Ethical and Societal Implications of GenAI
- Data Privacy and Security in GenAI Applications
- Guest Speaker: Security in GenAI Implementation
- Emerging Trends and Advancements in GenAI and Related Technologies

Final Project Presentations & Wrap-up (2 weeks)

- Student presentations of final projects

Assessment:

- Class participation and discussions (20%)
- Final project (group) (30%)
- Final exam (50%)

Class Participation

- Attendance and participation during class
- Assignment

Group Project

The main purpose of the group project is to simulate solving real-world business problems using Generative AI. By leveraging the key concepts from the lectures, students need to:

- Outline the nature of the business problem.
- Choose and explain the rationale for selecting a specific model to address that problem.
- Showcase the expected outcomes of the GenAI application and the success criteria.
- Explain the ongoing monitoring process and identify potential areas for improvement.
- Prepare a 15-minute presentation as a group.

Each group should have 5 to 7 students (peer evaluation to ensure fair contribution to the project)

Final Exam

- Multiple choice questions
- Short answer questions
- Essay questions

A careful review of lecture notes and reference materials is required for success in the exam.

- Explain key concepts covered in the lectures
- Case study questions on how to apply concepts/frameworks